

Republic of Yemen
Sana'a University
Postgraduate and
Scientific Research
Faculty of Dentistry
Department of Orthodontics,
Pedodontics and Prevention



**Closure and Stability of Diastema Associated With Generalized
Interdental Spacing Using Clear Aligners versus Fixed Appliances
with Adjunctive Frenectomy: A 2×2 Factorial RCT**

Research Proposal presented by:

Seham Abdulghani Humoud Al-Okam

B.D.S., Faculty of Dentistry, Ibb University, republic of Yemen, 2013

M.Sc., Kyrgyz State Medical Institute for Postgraduate Studies, 2017

Research Proposal submitted to

**Department of Orthodontics, Pedodontics and Prevention,
Faculty of Dentistry, Sana'a University, republic of Yemen**

For fulfillment of the requirements for the degree of

**Doctorate of Philosophy (PhD)
(Clinical Dental Science, Orthodontics)**

Under the supervision of

Prof. Dr. Ghamdan Al-Harazi

Professor, Department of Orthodontics, Pedodontics and Prevention,
Faculty of Dentistry, Sana'a University

(2026^{AD} – 1447^H)



الجمهورية اليمنية
جامعة صنعاء
نيابة الدراسات العليا والبحوث
كلية طب الأسنان
قسم تقويم الأسنان، والأطفال، والصحة الوقائية

إغلاق وثبات الدياستيما المصاحبة للفراغات بين الأسنان باستخدام التقويم الشفاف
مقارنةً بالتقويم الثابت مع قص اللجام كإجراء مساعد: تجربة سريرية عشوائية 2x2

مقترح بحث مقدم من

سهام عبدالغني حمود العكام

بكالوريوس طب وجراحة الأسنان، كلية طب الأسنان، جامعة إب، الجمهورية اليمنية (2013)
ماجستير، المعهد الطبي الحكومي للدراسات العليا في قرغيزستان (2017)

مقترح البحث مُقدّم إلى

قسم تقويم الأسنان وطب أسنان الأطفال والوقائية، كلية طب الأسنان، جامعة صنعاء،
الجمهورية اليمنية

استكمالاً لمتطلبات الحصول على

درجة دكتوراه الفلسفة

(العلوم السريرية لطب الأسنان، تقويم الأسنان)

تحت إشراف

د. غمدان الحرازي

بروفيسور، قسم تقويم الأسنان وطب أسنان الأطفال والوقائية
كلية طب الأسنان، جامعة صنعاء

(2026^{AD} – 1447^H)

Table of Contents

Section	Page
List of Abbreviations	1
Abstract	2
1. Introduction	3
2. Problem Statement	3
3. Justification	4
4. Research Hypotheses	4
5. Evidence from Previous Studies	5
6. Literature Review	6
7. Research Objectives	8
8. Materials and Methods	9
9. Recommendation	13
10. Timeline	14
11. Ethical Considerations	14
11. References	15
12. Appendices	34
Appendix A: Informed Consent Form (Bilingual)	16
Appendix B: Dental Examination and Case Documentation Form	17
Appendix C: Patient Wear Log and Satisfaction Form	20

List of Abbreviations

Abbreviation	Full Term
AJODO	American Journal of Orthodontics and Dentofacial Orthopedics
DoF	Depth of Field
IRB	Institutional Review Board
MBT	McLaughlin, Bennett, and Trevisi Prescription
mm	Millimeter
NiTi	Nickel–Titanium
RCT	Randomized Clinical Trial
SPSS	Statistical Package for the Social Sciences
SS	Stainless Steel
T0	Baseline (Pre-treatment)
T1	End of Active Orthodontic Treatment
T2	End of Retention Period
T3	6 Months Post-Retention
T4	8 Months Post-Retention
µm	Micrometer

Closure and Stability of Diastema Associated With Generalized Interdental Spacing Using Clear Aligners versus Fixed Appliances with Adjunctive Frenectomy: A 2×2 Factorial RCT

ABSTRACT

Diastema associated with generalized interdental spacing is a common orthodontic condition that affects dental esthetics and treatment stability, particularly when the diastema is ≥ 3 mm. Achieving stable space closure in such cases remains challenging, and relapse is a frequent concern.

Clear aligners and fixed orthodontic appliances are commonly used to manage diastema and generalized spacing. Fixed appliances provide reliable control of tooth movement, while clear aligners offer improved esthetics and patient comfort. However, evidence comparing their effectiveness and long-term stability in cases of generalized interdental spacing is limited.

Adjunctive frenectomy, especially when performed using laser technology, has been proposed to enhance the stability of diastema closure. Nevertheless, its role remains controversial, and high-quality randomized clinical trials evaluating the combined effects of appliance type and frenectomy are lacking.

Therefore, a prospective 2×2 factorial randomized clinical trial is needed to compare clear aligners and fixed appliances, with and without adjunctive laser frenectomy, in the management of diastema associated with generalized interdental spacing.

Key words: Diastema; Generalized interdental spacing; Clear aligners; Fixed orthodontic appliances; Laser frenectomy; Treatment stability; Randomized clinical trial

1. Introduction

Diastema associated with generalized interdental spacing is a frequent orthodontic condition that poses both esthetic and stability-related challenges. When multiple interdental spaces are present and the diastema measures ≥ 3 mm, achieving predictable and stable space closure becomes more complex, and the risk of post-treatment relapse increases.

Clear aligners and fixed orthodontic appliances are commonly used for the management of diastema and generalized spacing. Fixed appliances have traditionally been considered effective for space closure due to their ability to provide precise tooth movement. In recent years, clear aligners have gained popularity because of their esthetic advantages and improved patient comfort. However, their effectiveness and long-term stability in cases of generalized interdental spacing remain insufficiently documented.

Adjunctive frenectomy has been suggested as a supportive procedure to improve the stability of diastema closure, particularly in patients with abnormal labial frenum attachment. Laser-assisted frenectomy offers several clinical advantages, yet its role as an adjunct to orthodontic treatment in generalized spacing cases remains controversial.

Given the limited availability of high-quality randomized clinical trials addressing these factors simultaneously, this study aims to compare the closure efficiency and post-treatment stability of diastema associated with generalized interdental spacing treated with clear aligners and fixed appliances, with and without adjunctive laser frenectomy, using a 2×2 factorial randomized clinical trial design.

2. Problem Statement

Despite the widespread use of clear aligners and fixed orthodontic appliances for the treatment of diastema associated with generalized interdental spacing, there is limited high-quality evidence regarding their comparative effectiveness and long-term stability. In cases where diastema is ≥ 3 mm and accompanied by generalized spacing, achieving stable space closure remains challenging, and post-treatment relapse is common.

Additionally, the role of adjunctive laser frenectomy in enhancing the stability of diastema closure is not clearly established, with existing studies showing inconsistent findings and relying mainly on retrospective or non-randomized designs. The lack of prospective randomized clinical trials that simultaneously evaluate the effects of orthodontic appliance type and adjunctive frenectomy represents a significant gap in the current orthodontic literature.

Therefore, there is a need for a well-designed 2×2 factorial randomized clinical trial to compare clear aligners and fixed appliances, with and without adjunctive laser frenectomy, in the management of diastema associated with generalized interdental spacing, in order to provide evidence-based guidance for clinical decision-making and improve long-term treatment outcomes.

3. Justification

Diastema associated with generalized interdental spacing represents a common clinical problem with a high risk of post-treatment relapse, particularly when the diastema is ≥ 3 mm. Despite the increasing use of clear aligners, there is insufficient prospective evidence comparing their effectiveness and stability with conventional fixed appliances in such cases.

Furthermore, the role of adjunctive laser frenectomy in improving the stability of diastema closure remains controversial, and clinical decision-making is often based on individual experience rather than evidence from randomized trials. The absence of well-designed factorial randomized clinical trials evaluating both orthodontic appliance type and adjunctive frenectomy highlights a significant gap in the current literature.

This study is justified by its 2×2 factorial randomized clinical trial design, which allows simultaneous evaluation of treatment modality and frenectomy as independent and interacting factors. The findings are expected to provide clinically relevant evidence to optimize treatment planning, enhance long-term stability, and support evidence-based orthodontic practice.

3. Research Hypotheses

3.1. Primary Hypotheses

There is a significant difference in the closure efficiency and post-treatment stability of diastema associated with generalized interdental spacing between patients treated with clear aligners and those treated with fixed orthodontic appliances.

Adjunctive laser frenectomy significantly improves the post-treatment stability of diastema closure in patients with generalized interdental spacing compared to orthodontic treatment without frenectomy.

3.2. Secondary Hypotheses

The interaction between the type of orthodontic appliance and adjunctive laser frenectomy has a significant effect on the stability of diastema closure.

Patients treated with clear aligners with adjunctive laser frenectomy demonstrate comparable or improved stability of diastema closure compared to those treated with fixed appliances without frenectomy.

3.3. Null Hypotheses (H_0)

There is no significant difference in diastema closure or post-treatment stability between clear aligners and fixed orthodontic appliances.

Adjunctive laser frenectomy has no significant effect on the stability of diastema closure in patients with generalized interdental spacing.

There is no significant interaction between orthodontic appliance type and adjunctive laser frenectomy on treatment outcomes.

5. Evidence from Previous Studies:

***Title: Stability of Maxillary Interincisor Diastema Closure After Extraction Orthodontic Treatment**

Authors: M. J. Carruitero

Journal: *The Angle Orthodontist* -2020- Q1 Journal

Summary of the Study:

This study evaluated the long-term stability of maxillary interincisor diastema closure following orthodontic treatment involving first premolar extractions. A sample of 24 patients presenting with pre-treatment interincisor diastema was examined using dental casts and radiographs at three stages: pretreatment (T1), post-treatment (T2), and post-treatment follow-up (T3). Diastema relapse was defined as a reopening of 0.50 mm or greater.

The results showed no statistically significant relapse for maxillary interincisor diastemas. Clinically significant relapse occurred in 27.78% of cases, while the relapse specifically affecting the midline diastema was only 8.33%. Overall clinical stability of interincisor closing was 72.22%, and midline diastema stability reached 91.67%.

The study concluded that diastema closure after extraction-based orthodontic treatment is generally stable, with minimal relapse over time. Moreover, relapse was not significantly associated with diastema width, overjet, overbite, or angulation of maxillary anterior teeth. These findings support the effectiveness of extraction protocols in achieving stable diastema closure when appropriately managed.

***Title: Esthetics and Patient Level Evaluation of Fixed Orthodontic Treatment Conducted before and After Surgical Frenectomy Procedure for Midline Diastema Cases: An Original Research Study**

Authors: Ruchi Saini

Journal: *Journal of Chemical Health Risks (JCHR)*, Volume 14, Issue 6, Pages 46–53

Year of Publication: 2024 - Q4

Summary of the Study:

The study aimed to evaluate esthetic outcomes and patient-related factors following fixed orthodontic treatment performed **before** and **after** surgical frenectomy in cases of maxillary midline diastema. A total of 40 patients were selected through systematic sampling and were divided into two equal groups.

Group 1 underwent frenectomy **after** orthodontic space closure, while Group 2 received frenectomy **before** space closure.

Patient-centered outcomes such as smile esthetics, comfort, satisfaction, and social acceptance were assessed at two follow-up points: 2 months and 4 months. Results demonstrated that Group 1 consistently showed higher satisfaction levels and better esthetic scores compared with Group 2. Statistical analysis using SPSS revealed significant differences between and within groups ($p < 0.05$), with Group 1 exhibiting superior outcomes ($p = 0.001$).

Conclusion:

The study concluded that performing surgical frenectomy **after** orthodontic space closure provides better esthetic results and higher patient satisfaction. Frenectomy performed before space closure did not achieve optimal outcomes.

***Title: Prevalence and Etiology of Maxillary Midline Diastema among University Students, Hodeidah Governorate, Yemen: A Cross-sectional Study**

First Author: Al-Khawlani A.M.

Journal: *Journal of Contemporary Dental Practice (JCDP)*

Year of Publication: 2024

Journal Rank: Q2 (Scopus)

Summary of the Study:

This cross-sectional study aimed to assess the prevalence and etiological factors of maxillary midline diastema (MMD) among university students in Hodeidah Governorate, Yemen. A total of **1,661 students** aged 19–26 years were clinically examined to measure the presence and width of midline diastema. The study also recorded demographic variables, family history, and potential etiological factors such as high frenal attachment, spacing, and tooth morphology.

The results showed that the overall prevalence of MMD was **15.9%**, with a significantly higher rate in **males (17.9%)** compared to females (11.9%). A high labial frenum attachment was identified as the **most common etiology (54.3%)**, followed by generalized spacing. Nearly half of the affected participants reported a **positive family history** of diastema, indicating a possible genetic component. The most common diastema width was **0.5–1 mm**. The study concluded that maxillary midline diastema is a relatively common finding among Yemeni university students, with notable gender differences and a strong association with high frenal attachment. The authors emphasized the importance of understanding the underlying etiology to ensure accurate diagnosis and effective treatment planning.

***Title: Management of Midline Diastema with Aligners: Case Series**

Authors: Diptangshu Garai

Journal: *International Research Journal of Multidisciplinary Scope (IRJMS)*

Year: 2025,Q4

Summary:

This case series reports the management of maxillary midline diastema using clear aligners as a modern, esthetic alternative to conventional orthodontic appliances. The authors present three adult patients from different professional backgrounds who previously avoided fixed orthodontic treatment due to concerns about appearance, discomfort, and social embarrassment. Using aligners with a dedicated spacing protocol, all cases demonstrated effective closure of the midline diastema within approximately 12 months, with improved esthetics, hygiene, comfort, and patient acceptance.

The study highlights clear aligners as a viable modality for selected diastema cases, especially among adults seeking discreet orthodontic treatment. The authors emphasize the importance of proper diagnosis, case selection, and patient counseling to achieve predictable outcomes.

6. Literature Review

6.1. Diastema, generalized interdental spacing, and relapse risk

Diastema and generalized interdental spacing are commonly driven by a combination of tooth–arch size discrepancy, periodontal and occlusal factors, soft-tissue anatomy (especially the labial frenum), and post-treatment neuromuscular adaptation. A key clinical concern is **post-treatment relapse**, which is reported more frequently for **midline diastema** than for some lateral spaces. In a post-retention evaluation of orthodontically closed maxillary interincisor diastemas, **relapse was common**, and larger initial diastema width was associated

with higher relapse risk. [PMC](#)

More broadly, classic orthodontic stability literature highlights that long-term stability is multifactorial and variable, and that generalized spacing cases can show meaningful post-treatment changes without careful retention planning. journals.sagepub.com

6.2. Clear aligners versus fixed appliances for spacing closure

Fixed appliances are traditionally considered reliable for space closure due to continuous force systems and strong three-dimensional control. Clear aligners have grown rapidly in clinical use because of esthetics, comfort, and oral hygiene advantages. However, the literature continues to debate whether aligners provide the same level of control and “finish quality” as fixed appliances in more complex scenarios, and many reviews emphasize that the overall evidence base still needs stronger prospective trials in specific malocclusion categories, including challenging spacing patterns. [PMC+1](#)

Recent comparative and review papers often conclude that aligners can achieve comparable outcomes in selected cases, while fixed appliances may have advantages in complex tooth movements and detailed finishing, which can matter for **space closure stability**. [PMC+1](#)

Because your proposal targets **diastema associated with generalized interdental spacing** (not isolated minor spacing), this gap in high-quality, condition-specific comparative evidence supports the need for a randomized clinical trial.

6.3. The role of frenectomy and timing controversies

The labial frenum has long been discussed as a contributing factor to midline diastema and relapse, but the decision to perform frenectomy, and when, remains debated. A review on the association between the superior labial frenum and maxillary midline diastema notes that current literature generally recommends **addressing the diastema orthodontically first**, with frenectomy delayed in many cases to reduce relapse risk related to residual tension and to avoid unnecessary early surgery in developmental situations. [ScienceDirect](#)

In line with clinical practice trends, a survey of orthodontists reported that frenectomy is commonly preferred **after diastema closure**, and that retention is recommended regardless of whether frenectomy is performed. [PMC](#)

More recent orthodontic discussions continue to emphasize the importance of aligning frenectomy timing with the orthodontic plan and relapse prevention strategy, reinforcing why your study’s standardized approach (laser frenectomy only, with defined protocols) is valuable. [MDPI+1](#)

6.4. Retention strategies and post-retention stability measurement

Retention is central to diastema and generalized spacing management, because relapse can occur even after successful closure. Contemporary literature continues to compare fixed and removable retention approaches, underscoring that long-term stability often depends on retention design, patient compliance (for removable retention), and mechanical integrity (for bonded retainers). [MDPI](#)

In diastema cases specifically, long-term bonded retention is frequently discussed as a stability-supporting option, with clinical reports suggesting prolonged service life in many patients, although breakage risk remains a practical issue requiring monitoring. [Pocket Dentistry](#)

Given your protocol includes **6 months retention** and then **two stability checkpoints (6 and 8 months post-retention)**, the study can generate clinically meaningful data on

short-to-medium term stability after retention ends, which is a period when relapse may become apparent.

6.5. Why a 2×2 factorial RCT is needed for your topic

The main limitation across much of the existing literature is that appliance comparisons and frenectomy effects are often studied separately, using heterogeneous protocols, retrospective designs, or mixed case types. Your **2×2 factorial randomized clinical trial** directly addresses this gap by testing:

- **Appliance modality effect** (clear aligners vs fixed appliances)
- **Adjunctive frenectomy effect** (laser frenectomy vs no frenectomy)
- And the **interaction** between them (whether frenectomy benefits differ by appliance type)

This design is particularly relevant for generalized spacing patterns, where stability is influenced by multiple interacting factors (tooth movement control, soft-tissue tension, and retention). By standardizing key elements in your proposal, including **diastema ≥ 3 mm**, **laser frenectomy only**, and defined follow-up windows, the study is positioned to provide higher-quality, clinically actionable evidence compared with many prior reports.

Summary of the Literature Gap

Current evidence on the management of diastema associated with generalized interdental spacing is limited and largely based on retrospective or observational studies. Most published research focuses on isolated midline diastema or minor spacing and does not adequately assess post-retention stability.

Comparative data between clear aligners and fixed appliances in generalized spacing cases are scarce, and the long-term stability of these treatment modalities remains unclear. In addition, the role of adjunctive frenectomy is still controversial, with inconsistent findings and a lack of standardized surgical protocols.

Notably, there is a lack of prospective 2×2 factorial randomized clinical trials that simultaneously evaluate the effects of orthodontic appliance type and adjunctive laser frenectomy on the closure and stability of diastema associated with generalized interdental spacing. This gap justifies the need for the present study.

7. Research Objectives

7.1. Primary Objectives

1. To evaluate and compare the effectiveness of clear aligners and fixed orthodontic appliances in achieving closure of diastema associated with generalized interdental spacing.
2. To assess and compare the post-treatment and post-retention stability of diastema closure between clear aligners and fixed appliances.

7.2. Secondary Objectives

3. To determine the effect of adjunctive laser frenectomy on the stability of diastema closure in patients with generalized interdental spacing.
4. To evaluate the interaction between orthodontic appliance type and adjunctive laser frenectomy on diastema closure and stability.
5. To assess relapse rates at 6 and 8 months after completion of the retention period.

8. Materials and Methods

8.1. Study Design

This study will be conducted as a **prospective 2×2 factorial randomized clinical trial** designed to evaluate the effects of orthodontic appliance type and adjunctive laser frenectomy on the closure and stability of diastema associated with generalized interdental spacing.

8.2. Study Setting and Recruitment

Participants will be recruited from **one primary source:**
The Orthodontic Clinics, Faculty of Dentistry, Sana'a University, Yemen.

8.3. Study Sample and Sample Size Calculation

The sample size was calculated based on the primary outcome variable, which is the amount of diastema relapse (measured in millimeters) following orthodontic treatment and retention. The study is designed as a **2×2 factorial randomized clinical trial**, allowing evaluation of the main effects of orthodontic appliance type (clear aligners vs fixed appliances) and adjunctive laser frenectomy (with vs without frenectomy).

Sample size estimation was performed for the **main effects** of the factorial design. Patients were therefore grouped as follows for primary comparisons:

- Clear aligners vs fixed appliances (24 participants per level)
- Frenectomy vs no frenectomy (24 participants per level)

Assuming a **clinically meaningful difference (Δ) of 0.6 mm** in diastema relapse between groups, with an expected **standard deviation (SD) of 0.7 mm**, a **two-sided significance level (α) of 0.05**, and a **statistical power of 80%**, the required sample size was calculated using the standard formula for comparison of two means.

Based on these assumptions, the minimum required sample size was approximately **22 participants per level of the main effect**. To ensure adequate power and account for potential withdrawals, the total sample size was increased to **48 participants**, distributed equally into **four groups of 12 patients each**.

An anticipated dropout rate of **two participants per group** was considered during recruitment to maintain sufficient statistical power for the primary analyses.

8.4. Eligibility Criteria

-Inclusion Criteria

- Diastema associated with generalized interdental spacing
- Diastema width ≥ 3 mm
- Indication for orthodontic treatment
- Good general and periodontal health
- Ability to comply with treatment and follow-up
- Signed informed consent

-Exclusion Criteria

- Previous orthodontic treatment
- Craniofacial anomalies or syndromes
- Active periodontal disease
- Systemic conditions affecting bone metabolism
- Need for surgical procedures other than frenectomy

8.5. Group Allocation

All participants will present with **diastema associated with generalized interdental spacing (≥ 3 mm)** and will be randomized into four groups:

- **Group 1 (Aligner + Frenectomy):**
Clear aligner therapy **for diastema associated with generalized interdental spacing, with adjunctive laser frenectomy.**
- **Group 2 (Aligner only):**
Clear aligner therapy **for diastema associated with generalized interdental spacing, without frenectomy.**
- **Group 3 (Fixed + Frenectomy):**
Fixed orthodontic appliance therapy **for diastema associated with generalized interdental spacing, with adjunctive laser frenectomy.**
- **Group 4 (Fixed only):**
Fixed orthodontic appliance therapy **for diastema associated with generalized interdental spacing, without frenectomy.**

Adjunctive frenectomy will be performed using **laser technique only** following a standardized protocol.

8.6. Treatment Protocol

A. Clear Aligner Treatment Protocol

1. **Initial Assessment**
Comprehensive orthodontic examination including intraoral and extraoral assessment, occlusal analysis, periodontal evaluation, and baseline measurement of diastema and generalized interdental spacing.
2. **Records and Digital Workflow**
Digital dental impressions will be obtained using an **intraoral scanner (Zahndent)**

with an accuracy of **15 microns** and a **depth of field of 15 mm**. Standardized intraoral and extraoral photographs will also be recorded.

3. **Digital Treatment Planning**

A virtual setup will be created to achieve sequential space closure with controlled bodily tooth movement. Attachments and staging will be planned to ensure optimal force delivery and root control. The final plan will be approved by the principal investigator.

4. **Aligner Fabrication and Delivery**

Clear aligners will be manufactured by **Dentalign Company, Sana'a Governorate, Yemen**. Patients will be instructed to wear aligners **20–22 hours per day**.

5. **Auxiliary Mechanics (Bodily Movement Control)**

Buttons will be bonded in the **cervical region** of selected teeth in both aligner and fixed appliance groups to enhance control of **bodily tooth movement** during space closure.

6. **Monitoring and Refinement**

Patients will be reviewed periodically to assess aligner fit, compliance, and progress. Refinement aligners will be prescribed when necessary.

7. **Completion of Active Treatment**

Active treatment will be completed when complete diastema closure and satisfactory alignment are achieved.

B. Fixed Orthodontic Appliance Treatment Protocol

1. **Appliance System**

All patients in the fixed appliance groups will be treated using **MBT prescription brackets with a 0.022-inch slot**.

2. **Archwire Sequence**

The standardized archwire sequence will include:

- Initial leveling and alignment wires: **0.012, 0.014, 0.016, and 0.018 NiTi**
- Rectangular stainless steel wires for space closure and torque control: **0.016 × 0.022, 0.017 × 0.025, and 0.019 × 0.025**

3. **Space Closure Mechanics**

Diastema and generalized interdental spaces will be closed using controlled mechanics aimed at achieving **bodily tooth movement** and maintaining root parallelism.

4. **Auxiliary Mechanics (Buttons)**

Buttons will be bonded in the **cervical region** of selected teeth, similar to the clear aligner groups, to enhance bodily movement control.

5. **Monitoring and Completion**

Patients will be reviewed at regular intervals. Active treatment will be completed once complete space closure and acceptable occlusal results are achieved.

8.7. Adjunctive Laser Frenectomy Protocol

For patients assigned to frenectomy groups, frenectomy will be performed using **laser technology only** after partial or complete space closure. Healing will be monitored clinically, and standardized post-operative instructions will be provided.

8.8. Retention Protocol

All patients will receive a **standardized retention protocol for 6 months** following completion of active orthodontic treatment.

8.9. Post-Retention Stability Assessment

Sixteen patients (four from each group) will be randomly selected:

- **8 patients** evaluated **6 months after retention**
- **8 patients** evaluated **8 months after retention**

8.10. Measurement Points

-Clinical and Digital Measurements

Measurements will be recorded at the following time points:

- **T0:** Baseline (pre-treatment)
- **T1:** End of active orthodontic treatment
- **T2:** End of retention period (6 months)
- **T3:** 6 months post-retention
- **T4:** 8 months post-retention

The following parameters will be assessed:

- Diastema width (mm)
- Amount of space closure
- Relapse (reopening of spaces)
- Root parallelism and bodily tooth movement

8.11. Radiographic Assessment

Periapical radiographs using a **digital periapical sensor** will be obtained at baseline and after space closure to confirm **bodily tooth movement** and assess root positioning.

8.12. Statistical Analysis

Data will be analyzed using statistical software (SPSS or equivalent). Descriptive statistics, including mean, standard deviation, median, and range, will be calculated for all quantitative variables.

Normality of data distribution will be assessed using the **Shapiro–Wilk test**. For normally distributed data, parametric tests will be applied, while non-parametric tests will be used when normality assumptions are not met.

Because the study is designed as a **2×2 factorial randomized clinical trial**, the primary analysis will be performed using **two-way factorial analysis** to evaluate:

- The main effect of orthodontic appliance type (clear aligners vs fixed appliances)

- The main effect of adjunctive laser frenectomy (with vs without frenectomy)
- The interaction effect between appliance type and frenectomy on treatment outcomes

Intergroup comparisons at different time points (T0–T4) will be conducted using appropriate tests, such as repeated-measures analysis for within-group changes and independent comparisons for between-group differences.

Primary outcome variables will include the amount of diastema closure and post-treatment stability. Secondary outcomes will include relapse rate and radiographic assessment of bodily tooth movement.

The level of statistical significance will be set at $p < 0.05$ for all tests. Missing data resulting from patient withdrawal will be handled according to standard statistical procedures appropriate for clinical trials.

8.12. Funding

This study will be **self-funded by the principal investigator**, with no external financial support.

9. Recommendations

Based on the design and objectives of the present study, the following recommendations are proposed:

1. Clear aligners and fixed orthodontic appliances should be carefully selected for the management of diastema associated with generalized interdental spacing, taking into account both treatment efficiency and long-term stability.
2. Adjunctive laser frenectomy may be considered as a supportive procedure in selected cases of diastema with generalized interdental spacing to enhance post-treatment stability, especially when an abnormal labial frenum is present.
3. Standardized treatment protocols, including controlled bodily tooth movement and appropriate retention strategies, are essential to minimize relapse following diastema closure.
4. Long-term follow-up beyond the retention period is recommended to better evaluate the stability of orthodontic treatment outcomes in generalized spacing cases.
5. Further multicenter randomized clinical trials with larger sample sizes and extended follow-up periods are recommended to confirm the findings of this study and improve the generalizability of the results.

10. Timeline

The study will be conducted over approximately 30 months:

Phase	Activity	Duration (Months)
Phase 1	Finalization of research protocol, ethical approval, ClinicalTrials.gov registration, calibration of measurements	Months 1–3
Phase 2	Patient recruitment, informed consent, baseline records, randomization and group allocation	Months 4–9
Phase 3	Active orthodontic treatment (clear aligners or fixed appliances) and adjunctive laser frenectomy (when indicated)	Months 10–21
Phase 4	Retention phase (standardized retention protocol for all participants)	Months 22–27
Phase 5	Post-retention stability assessment at 6 and 8 months (selected patients)	Months 28–29
Phase 6	Data analysis, interpretation of results, thesis writing, and manuscript preparation	Month 30

11. Ethical Considerations

This study will be conducted in accordance with the ethical principles outlined in the Declaration of Helsinki and relevant institutional guidelines. Ethical approval will be obtained from the Research Ethics Committee of the Faculty of Dentistry, Sana'a University, Yemen, prior to the commencement of the study.

All participants will receive a clear explanation of the study objectives, procedures, potential benefits, and possible risks before enrollment. Written informed consent will be obtained from all participants prior to participation. Participants will be informed of their right to withdraw from the study at any time without any effect on their ongoing or future treatment.

Patient confidentiality will be strictly maintained. All collected data will be anonymized and coded to ensure privacy, and access to study data will be restricted to the research team only.

Adjunctive laser frenectomy will be performed only when indicated and according to standardized clinical protocols to minimize risk. All clinical procedures used in this study are routinely performed in orthodontic practice and do not expose participants to undue risk beyond standard care.

The study will be self-funded by the principal investigator, and no external financial support or conflicts of interest are declared.

12. References

1. Edwards JG.
The diastema, the frenum, the frenectomy: A clinical study.
Am J Orthod. 1977;71(5):489–508.
2. Shashua D, Artun J.
Relapse after orthodontic correction of maxillary median diastema: A follow-up evaluation of consecutive cases.
Angle Orthod. 1999;69(3):257–263.
3. Huang WJ, Creath CJ.
The midline diastema: A review of its etiology and treatment.
Pediatr Dent. 1995;17(3):171–179.
4. Campbell PM, Moore JW, Matthews JL.
Orthodontically corrected midline diastemas: A histologic study and surgical procedure.
Am J Orthod. 1975;67(2):139–158.
5. Riedel RA.
Esthetics and its relation to orthodontic therapy.
Angle Orthod. 1950;20(3):168–178.
6. Little RM, Riedel RA, Stein A.
Mandibular arch length increase during the mixed dentition: Postretention evaluation of stability and relapse.
Am J Orthod Dentofacial Orthop. 1990;97(5):393–404.
7. Rossini G, Parrini S, Castrolforio T, Deregibus A, Debernardi CL.
Efficacy of clear aligners in controlling orthodontic tooth movement: A systematic review.
Angle Orthod. 2015;85(5):881–889.
8. Papadimitriou A, Mousouleas S, Gkantidis N, Kloukos D.
Clinical effectiveness of Invisalign orthodontic treatment: A systematic review.
Prog Orthod. 2018;19(1):37.
9. Buschang PH, Shaw SG, Ross M, Crosby D, Campbell PM.
Comparative time efficiency of aligner therapy and conventional edgewise braces.
Angle Orthod. 2014;84(3):391–396.
10. Janson G, Valarelli FP, Henriques JFC, de Freitas MR, Cançado RH.
Stability of anterior open-bite treatment with fixed appliances.
Am J Orthod Dentofacial Orthop. 2003;124(3):265–276.
11. Moyers RE.
Handbook of Orthodontics. 4th ed. Chicago: Year Book Medical Publishers; 1988.
12. Proffit WR, Fields HW, Larson B, Sarver DM.
Contemporary Orthodontics. 6th ed. St Louis: Elsevier; 2019.
13. Tanaka OM, Vitral RWF, Tanaka GY, Guerrero AP, Camargo ES.
Nonsurgical treatment of midline diastema using orthodontic movement.
Dental Press J Orthod. 2010;15(6):124–131.
14. Yared KF, Zenobio EG, Pacheco W.
Periodontal status of mandibular central incisors after orthodontic closure of diastema.
Am J Orthod Dentofacial Orthop. 2006;130(1):1–8.
15. Artun J, Spadafora AT, Shapiro PA.
A 3-year follow-up study of various types of orthodontic canine-to-canine retainers.
Eur J Orthod. 1997;19(5):501–509.

11. Appendices

Appendix A: Informed Consent Form (Bilingual)

نموذج الموافقة المستنيرة

Informed Consent Form

أوافق على المشاركة في هذه الدراسة بعد أن تم شرح أهدافها ومخاطرها المحتملة لي بشكل واضح.

I agree to participate in this study after its objectives and potential risks were clearly explained to me.

أفهم أن مشاركتي طوعية بالكامل، وأن بياناتي ستظل سرية ولن تُستخدم إلا لأغراض البحث العلمي.

I understand that my participation is completely voluntary, and my data will be kept confidential and used only for scientific research purposes.

أعلم أن لي الحق في الانسحاب من الدراسة في أي وقت دون إبداء أي أسباب، ودون أن يؤثر ذلك على الرعاية الطبية التي أتلقاها.

I know that I have the right to withdraw from the study at any time without giving any reason, and without this affecting the medical care I receive.

توقيع المشارك:

Participant Signature:

التاريخ:

Date:

توقيع الباحث:

Investigator Signature:

التاريخ:

Date:

Dental Examination and Case Documentation Form

Section A: Patient Identification

- Patient Code: _____
 - Age (years): _____
 - Sex: ☐ Male ☐ Female
 - Date of Examination: _____
 - Clinic: Orthodontic Clinics, Faculty of Dentistry, Sana'a University
-

Section B: Medical and Dental History

- Relevant medical history: ☐ Yes ☐ No
If yes, specify: _____
 - Medications: ☐ Yes ☐ No
If yes, specify: _____
 - Previous dental treatment: ☐ Yes ☐ No
 - Previous orthodontic treatment: ☐ Yes ☐ No
-

Section C: Clinical Examination

- Oral hygiene status: ☐ Good ☐ Fair ☐ Poor
 - Periodontal status: ☐ Healthy ☐ Gingivitis ☐ Periodontitis
 - Labial frenum type: ☐ Normal ☐ Thick ☐ High attachment
 - Functional habits (if any): _____
-

Section D: Occlusal Assessment

- Angle classification: ☐ Class I ☐ Class II ☐ Class III
 - Overjet (mm): _____
 - Overbite (% or mm): _____
 - Presence of generalized interdental spacing: ☐ Yes ☐ No
-

Section E: Diastema Assessment

- Location: ☐ Maxillary ☐ Mandibular ☐ Both
- Midline diastema width (mm): _____
- Total interdental spacing (mm): _____
- Diastema ≥ 3 mm: ☐ Yes ☐ No

Section F: Diagnostic Records

- Intraoral photographs: ☐ Taken ☐ Not taken
 - Extraoral photographs: ☐ Taken ☐ Not taken
 - Intraoral scan (Zahndent): ☐ Completed ☐ Pending
 - Periapical radiographs: ☐ Taken ☐ Not taken
-

Section G: Treatment Allocation

- Appliance type:
 - ☐ Clear aligners (Dentalign)
 - ☐ Fixed appliances (MBT 0.022)
 - Frenectomy:
 - ☐ Laser frenectomy
 - ☐ No frenectomy
 - Group assignment: ☐ G1 ☐ G2 ☐ G3 ☐ G4
-

Section H: Treatment Progress (Follow-up)

Visit	Date	Aligner/Wire Stage	Space Closure Progress	Notes
V1				
V2				
V3				

Section I: Retention Phase

- Retention start date: _____
- Retainer type: _____
- Retention compliance: ☐ Good ☐ Fair ☐ Poor

Section J: Stability Evaluation

- Evaluation time point: ☐ T2 ☐ T3 ☐ T4
 - Diastema width (mm): _____
 - Relapse present: ☐ Yes ☐ No
 - Root parallelism confirmed radiographically: ☐ Yes ☐ No
-

Section K: Examiner Details

- Examiner name: _____
- Signature: _____
- Date: _____

Patient Wear Log and Satisfaction Form

Section A: Patient Information

- Patient Code: _____
 - Age: _____
 - Sex: ☐ Male ☐ Female
 - Treatment group:
 - ☐ Clear aligners
 - ☐ Fixed appliances
-

Section B: Appliance Wear Log

(For Clear Aligner Patients)

1. Average daily aligner wear time:
 - ☐ < 18 hours/day
 - ☐ 18–20 hours/day
 - ☐ 20–22 hours/day
 2. Aligner wear compliance:
 - ☐ Excellent
 - ☐ Good
 - ☐ Fair
 - ☐ Poor
 3. Frequency of aligner removal outside meals:
 - ☐ Rarely
 - ☐ Sometimes
 - ☐ Often
 4. Any difficulties wearing aligners?
 - ☐ Yes
 - ☐ NoIf yes, specify: _____
-

(For Fixed Appliance Patients)

5. Appliance breakage or debonding during treatment:
 - ☐ Yes
 - ☐ No
6. Missed appointments:
 - ☐ None
 - ☐ 1–2 visits
 - ☐ > 2 visits
 - ☐

7. Compliance with oral hygiene instructions:

- ☐ Excellent
 - ☐ Good
 - ☐ Fair
 - ☐ Poor
-

Section C: Comfort and Pain Assessment

8. Level of discomfort during the first week of treatment:

- ☐ None
- ☐ Mild
- ☐ Moderate
- ☐ Severe

9. Pain during space closure phase:

- ☐ None
- ☐ Mild
- ☐ Moderate
- ☐ Severe

10. Use of pain medication:

- ☐ Yes
 - ☐ No
-

Section D: Esthetic and Functional Satisfaction

11. Satisfaction with dental appearance during treatment:

- ☐ Very satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied

12. Satisfaction with speech during treatment:

- ☐ Very satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied

13. Ease of eating during treatment:

- ☐ Very easy
 - ☐ Easy
 - ☐ Moderate difficulty
 - ☐ Difficult
-

Section E: Overall Satisfaction

14. Overall satisfaction with orthodontic treatment:

- ☐ Very satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied

15. Would you choose the same treatment again?

- ☐ Yes
- ☐ No

16. Would you recommend this treatment to others?

- ☐ Yes
 - ☐ No
-

Section F: Retention Phase Feedback

17. Retainer wear compliance:

- ☐ Excellent
- ☐ Good
- ☐ Fair
- ☐ Poor

18. Comfort while wearing the retainer:

- ☐ Very comfortable
- ☐ Comfortable
- ☐ Uncomfortable

Section G: Additional Comments

19. Patient comments or suggestions:

Section H: Examiner Information

- Examiner name: _____
- Date: _____